

Inheritance:

Advantage: → Code reusability, method overriding
Easy maintenance, organized code

Types → Single Inheritance $\begin{matrix} P & C \\ A & \rightarrow B \end{matrix}$ Multilevel → $A \rightarrow B \rightarrow C$
Hierarchical $\begin{matrix} P & P & C \\ A \rightarrow B, & A \rightarrow C \end{matrix}$ Multiple $\begin{matrix} P \\ A, B \end{matrix} \rightarrow C$ $\begin{matrix} \text{Par} & \text{child} \end{matrix}$

Single Inheritance
Parent → child

Hybrid Inheritance → combination of above types

Multilevel Inheritance
Grandfather → Parent → child.

Hierarchical Inheritance →

Multiple Inheritance
Class A > Class C.
Class B

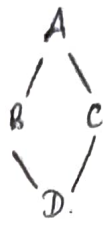
Parent (Animal)
├── child (Dog)
└── child (Cat)

Multiple Inheritance → not support with classes.

one class inherits from two (or) more classes

Hybrid Inheritance → combination of two (or) more types of Inheritance

(Hierarchical + multiple)



[Java doesn't support Hybrid Inheritance directly through class but allows it via interfaces].

Super keyword

↳ refer to the immediate parent class object.

↳ It's a reference variable. also it's help the child to access the methods, variables & constructors of it's parent class.

When sub class has a variable with same name as it's parent class, it hides the parent one.

Super.variablename → use this keyword

[Super() used to call the parent class constructor from the child (must be first statement inside the child)]

Super.variable → access parent variable

super() → call parent constructor

super.method() → call parent method

Interface

[All interface methods are public & abstract by default]

In java An Interface is a Blueprint of a class.

It contains abstract methods (without body) that must be implement by the class.

Need of it → To achieve multiple inheritance
To achieve abstraction

↗ [Don't allow multiple inheritance but allow implement multiple interface]

→ [Hiding implementation details & show what needs to be done]

Final Keyword

It's a non-access modifier used to restrict changes to variables, methods & classes. [final method can't be overridden by subclasses]
Cannot be modified, overridden or inherited.
Value can't be changed once assigned (final variable)
[final class can't be inherit by any class]

Instanceof Keyword

To define the object, checks the relationship between an object & class.
Used to test whether an object is an instance of a specific class (or) sub class.
[return boolean value, used for type checking]

ObjectName instanceof className. → Syntax.

this Keyword

it refers to the current object of the class.

[Particular refer]